

MATH 102 Calculus II
Final exam (mock)

10 problems, 10 points each

Total: 100 points

13:30 – 16:30, December 12, 2025

1. (10 points) Let $D = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 \leq 4, y^2 + z^2 \leq 4\}$. Find the volume of D .
2. (10 points) Let D be the domain in $\mathbb{R}^2$ given by $9(x-1)^2 + 4(y-3)^2 \leq 36$. Find

$$\int_D (x+y) dA$$

3. (10 points) Let E be the shape of an 'ice cream cone' with its vertex at the origin $(0,0,0)$, its axis along the positive z -axis, a top spherical surface of radius R , and a cone angle of ϕ_0 . Find the average distance from any point inside this solid E to the origin $(0,0,0)$.
4. (10 points) Let $C \subseteq \mathbb{R}^2$ be the curve defined by

$$y^2 = x^3 + x^2$$

in the second and the third quadrant (i.e. the $-1 \leq x \leq 0$ region), oriented counter-clockwise. Find the work done by force $F = (2xy, -x^3)$ along C .

5. (10 points) Given the vector field $F(x, y) = (ye^{xy}, xe^{xy})$. The path C is closed that traces the boundary of the rectangle with vertices at $(0,0)$, $(1,0)$, $(1,2)$, and $(0,2)$ in the counterclockwise direction. Find the amount of work done by the vector field F along the path C .
6. (10 points) Let $F(x, y, z) = (x, y, z)$ be the vector field. Let $S = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 9, z \geq 0\}$. Derive the flux of the vector field on S .
7. (10 points) Let F be the vector field defined for $(x, y, z) \neq (0,0,0)$ by

$$F(x, y, z) = \frac{(x, y, z)}{(x^2 + y^2 + z^2)^{3/2}}$$

Prove that there is no vector field $\mathbf{G}$ such that $F = \nabla \times \mathbf{G}$.

8. (10 points) Let S be the surface formed by the part of the paraboloid $z = 1 - x^2 - y^2$ lying above the xy -plane, and let $F = (x, y, 2(1-z))$. Calculate the upward flux of F across S by the following two ways:
 - (a) By direct calculation of $\int_S F \cdot N d\sigma$.
 - (b) Using divergence theorem and computing the flux across a simpler surface.
9. (10 points) Let $F : \mathbb{R}^2 \setminus \{(0,0)\} \rightarrow \mathbb{R}^2$ be the vector field given by

$$F(x, y) = \left(\frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right)$$

Is it conservative? If so, find its potential. If not, explain why.

10. (10 points) Let S be a part of the surface of the cone

$$x = 1 - \sqrt{y^2 + z^2}, \quad 0 \leq x \leq 1, \quad z \geq 0$$

and let the vector field $F = (xz, xy^2 + 2z, xy + z)$. Suppose that S is oriented with a unit normal vector field N such that $N \cdot e_3 > 0$.

(a) Find $\int_S \operatorname{curl} F \cdot N d\sigma$.

(b) Find $\int_{\partial S} F \cdot T ds$, where ∂S is equipped with the induced orientation from S .

This is the end of the test.

Remember, this exam is just preparation for the real one, so please don't be discouraged even if you don't get the score you hoped for.

As you may know, today is the last day we meet. You all have done a great job this semester, and it was an absolute honor to meet every single one of you. Keep up the excellent work, and good luck on your final exams!